

5.1 Energy: Conservation, Work, Power & Efficiency

Question Paper

Course	CIE A Level Physics
Section	5. Work, Energy & Power
Topic	5.1 Energy: Conservation, Work, Power & Efficiency
Difficulty	Hard

Time allowed: 20

Score: /9

Percentage: /100

Question 1

A turbine at a hydroelectric power station is situated 30 m below the level of the surface of a large lake. The water passes through the turbine at a rate of 340 m^3 per minute.

The overall efficiency of the turbine and generator system is 90 %.

What is the output power of the power station? (The density of water is 1000 kg m^{-3})

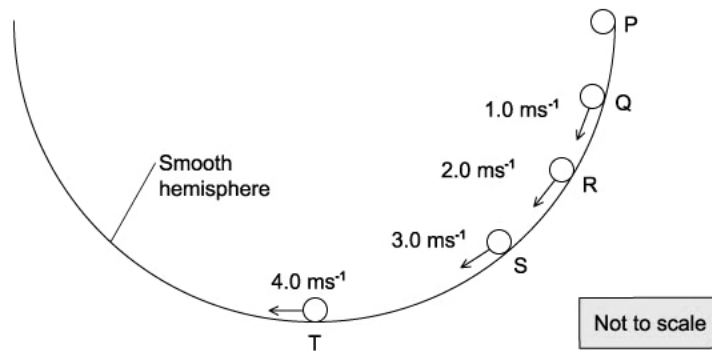
- A. 0.15 MW
- B. 1.5 MW
- C. 1.7 MW
- D. 90 MW

[1 mark]

Question 2

A small mass is placed at point P on the inside surface of a smooth hemisphere. It is then released from rest. When it reaches the lowest point T, its speed is 4.0 m s^{-1} .

The diagram (not to scale) shows the speed of the mass at other points Q, R and S as it slides down. Air resistance is negligible.



The mass loses potential energy E in falling from P to T.

At which point has the mass lost potential energy $\frac{E}{4}$?

- A. Q
- B. R
- C. S
- D. none of these

[1 mark]

Question 3

A wind turbine has blades that sweep an area of 2000 m^2 . It converts the power available in the wind to electrical power with an efficiency of 50 %.

What is the electrical power generated if the wind speed is 10 m s^{-1} ? (The density of air is 1.3 kg m^{-3})

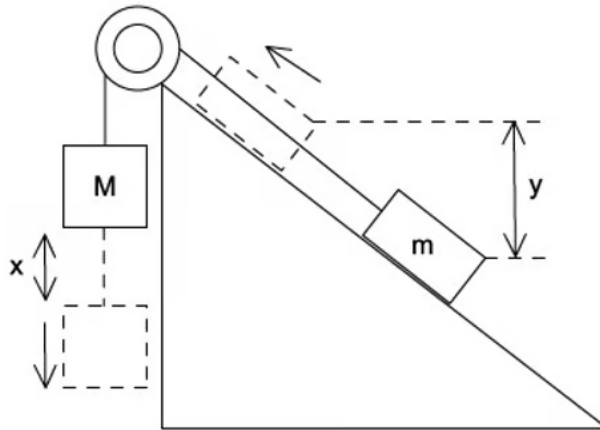
- A. 130 kW
- B. 650 kW
- C. 1300 kW
- D. 2600 kW

[1 mark]

Question 4

The figure shows a mass m on an inclined plane connected by an inextensible string over a smooth pulley to a larger mass M . When the mass M falls through a distance x , the mass m moves up a vertical distance y at a constant speed.

The friction between m and the inclined plane is F .



What amount of energy is converted to heat?

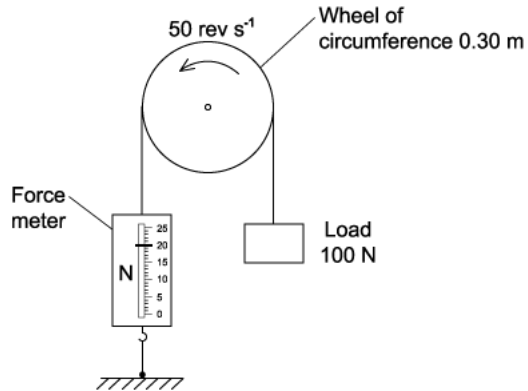
- A. $F(x - y)$
- B. Fy
- C. $Mgx - mgy$
- D. $(M - m)gx$

[1 mark]

Question 5

The diagram shows a wheel of circumference 0.30 m. A rope is fastened at one end to a force meter. The rope passes over the wheel and supports a freely hanging load of 100 N. The wheel is driven by an electric motor at a constant rate of 50 revolutions per second.

When the wheel is turning at this rate, the force meter reads 20 N.



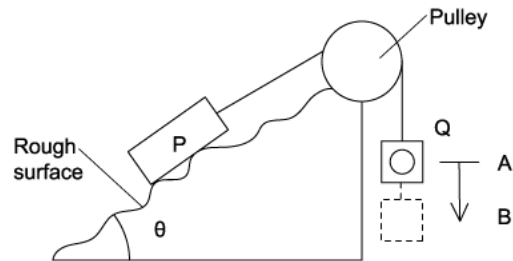
What is the output power of the motor?

- A. 0.3 kW
- B. 1.2 kW
- C. 1.8 kW
- D. 3.8 kW

[1 mark]

Question 6

The figure below shows two blocks P and Q, initially stationary and connected to each other by a taut string passing over a smooth pulley. The system is released from rest and block Q moves vertically downwards from position A to B.



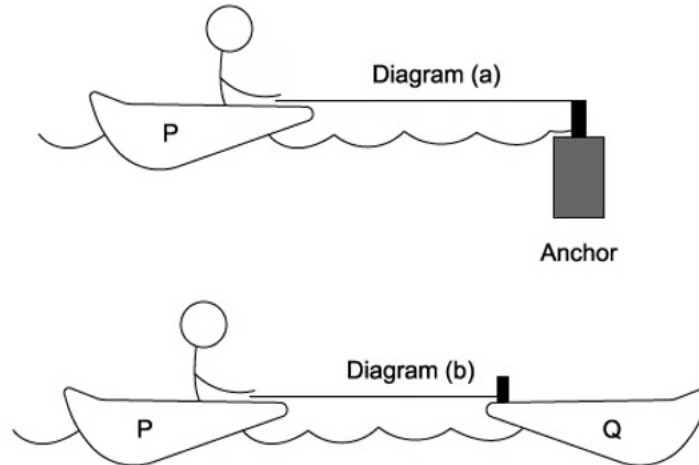
Which of the following statements is correct?

- A. Potential energy of the block Q at A = Kinetic energy of block Q at B
- B. Loss in potential energy of the whole system = Gain in kinetic energy of the whole system + work done against friction due to block P moving against a rough surface
- C. Loss in potential energy of the whole system = Gain in kinetic energy of the system
- D. Loss in potential energy of block Q = Gain in potential energy of block P

[1 mark]

Question 7

A boy on a boat P pulls on a rope with a constant force F over a duration of time t . The other end of the rope is either tied to an anchor on the pier (Diagram a) or a freely floating boat Q (Diagram b) of equal mass as shown in the figures shown.



W_a and W_b represent the total work done by the boy during the time t and P_a and P_b represent the average power output of the boy in diagrams (a) and (b) respectively.

Which of the following statements is correct?

- A. $W_a > W_b$ and $P_a > P_b$
- B. $W_a = W_b$ and $P_a = P_b$
- C. $W_a < W_b$ and $P_a < P_b$
- D. $W_a < W_b$ and $P_a = P_b$

[1 mark]

Question 8

What is the power required to give a body of mass m a forward acceleration a when it is moving with velocity v up a frictionless track inclined at an angle θ to the horizontal?

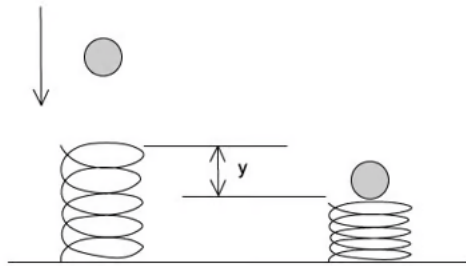
- A. $mav \sin \theta$
- B. $mav \sin \theta + mgv$
- C. $mav + mgv \sin \theta$
- D. $(mav + mgv) \sin \theta$

[1 mark]

Question 9

A ball of mass m falls freely from rest. When it has reached a speed v , it strikes a vertical spring.

The spring is compressed by a distance y before the ball moves upwards again.



Assume that all energy the ball loses becomes elastic potential energy in the spring.

What is the average force exerted by the spring during its compression?

- A. $\frac{mv^2}{2y}$
- B. $\frac{m}{2y}(v^2 - 2gy)$
- C. $\frac{mv^2}{y}$
- D. $\frac{m}{2y}(v^2 + 2gy)$

[1 mark]

Model Answers: Hard

1

The correct answer is **B** because:

- At the top of the lake, the water has gravitational potential energy as it is above the ground
- As it passes through the turbine, the energy is converted to electrical energy, however, the system is only 90% efficient, so, not all of the gravitational potential energy becomes electrical energy
 - GPE of water = mgh
 - Input Power = $\frac{\text{energy}}{\text{time}} = \frac{GPE}{\text{time}} = \frac{mgh}{t}$
- The water passes through the turbine at a rate of 340 m^3 per minute. This quantity represents the volume flow rate and has unit m^3 per unit or $\frac{\Delta V}{t}$
- Volume flow rate = $\frac{\Delta V}{t} = 340 \text{ m}^3 \text{ min}^{-1}$
- We need to convert this into SI unit
 - In 1 minute, a volume of 340 m^3 passes through the turbine
 - In 1 second, a volume of $\frac{340}{60} = 5.67 \text{ m}^3$ passes through the turbine
- So, the volume flow rate is $\frac{\Delta V}{t} = 5.67 \text{ m}^3 \text{ s}^{-1}$
- The formula for the input power above requires the mass, not the volume. We need to convert the volume flow rate into mass flow water or $\frac{\Delta m}{t}$
- Density is equal to $\rho = \frac{\text{mass}}{\text{volume}} = \frac{m}{V}$, so, $\Delta m = \rho V$
- We must divide by time to get the equivalent mass flow rate :
 - $\frac{\Delta m}{t} = \frac{\rho V}{t} = 1000 \times 5.67 = 5670 \text{ kg s}^{-1}$
 - Input Power = $\frac{mgh}{t} = \left(\frac{\Delta m}{t} \right) gh$

◦ Input Power $5670 \times 9.81 \times 30 = 1.67 \times 10^6 W = 1.67 MW$

- The overall efficiency of the system is 90%
- Therefore, output power = $0.90 \times 1.67 = 1.5 MW$

2

The correct answer is **B** because:

- From conservation of energy:
- The potential energy (PE) lost in falling from P to T = Kinetic energy (KE) of mass at T

◦ $E = \frac{1}{4} m(4)^2 = 8m$

- Since the mass is at rest at P, its KE is zero and its PE is at a maximum
- Since T is the lowest point, PE is zero here
- So, the total energy of the mass in the system is E
- In general, PE lost at a point = KE at that point
- From above, $E = 8m$, so, a PE loss of $\frac{E}{4} = 2m$
 - At R: $KE = \frac{1}{2} m(2)^2 = 2m = PE \text{ lost}$
- Therefore, when the mass is at R, it has half the final speed and so has gained a kinetic energy of $\frac{E}{4}$, this is when it has lost a potential energy of $\frac{E}{4}$

A is incorrect because At Q: $KE = \frac{1}{2} m(1)^2 = 0.5m = PE \text{ lost}$, this is a loss of $\frac{E}{16}$

C is incorrect because At S: $KE = \frac{1}{2} m(3)^2 = \frac{9}{2} = PE \text{ lost}$, this is a loss of $\frac{16E}{9}$

D is incorrect because At R: $KE = \frac{1}{2} m(2)^2 = 2m = PE \text{ lost}$, this is a loss of $\frac{E}{4}$

3

The correct answer is **B** because:

- The kinetic energy of the air causes the wind turbine to rotate which is then converted to electrical power

- Kinetic energy of air = $\frac{1}{2}mv^2$
- The speed is known so we need to find the mass using:
 - Density = $\frac{\text{mass}}{\text{volume}}$, so, mass = density $\times$ volume
 - Density of air = 1.3 kg m^{-3}
- The wind speed is 10 m s^{-1} and this wind blows on an area of 2000 m^2
- This means in one second, a column of air of 10 m in length and with an area of 2000 m^2 moves past the blades of the wind turbine
 - This is equivalent to a volume of $10 \times 2000 = 20\,000 \text{ m}^3 \text{ s}^{-1}$
- The mass of air passing the blades per second = density $\times$ volume of air per second
- Mass of air passing the blades per second: $\frac{m}{t} = 1.3 \times 20\,000 = 26\,000 \text{ kg s}^{-1}$
 - Power from wind = $\frac{\text{energy}}{\text{wind}} = \frac{mv^2}{2t} = \frac{1}{2} \left(\frac{m}{t} \right) v^2$
- Efficiency = 50%, so, only 50% of the power of the wind is converted to electrical power
 - Electrical Power = $0.50 \times (0.5 \times 26\,000 \times 10^2) = 650\,000 = 650 \text{ kW}$

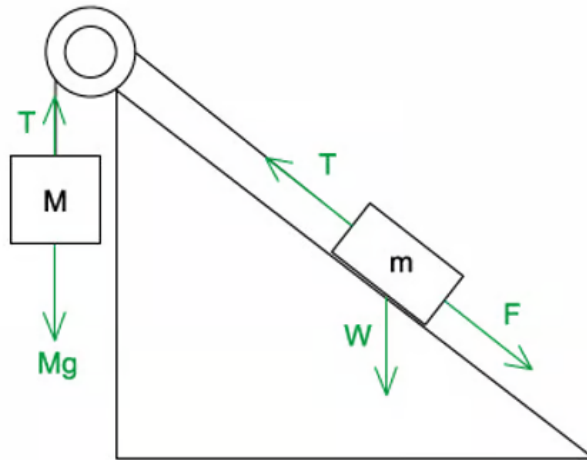
Therefore, the electrical power output of the turbine is 650 kW

4

The correct answer is **C** because:

- Work done = Force applied $\times$ Distance moved in the direction of force
- The question asks how much energy is converted to heat energy; this energy comes from the friction experienced by mass m moving up the rough plane
- In this case the forces acting on the mass are its weight W , the frictional force F and the tension T in the string holding the mass which is equal to the weight of mass M , so we can write
 - Vertical force on mass M : $T = Mg$
 - Weight of mass m : $W = mg$
 - Vertical forces on mass m : $F + W = T$
 - Therefore, the frictional force is: $F = T - W = Mg - mg$
- Vertical distance travelled by mass $M = x$
- Work done by mass $M = Mgx$
- Vertical distance travelled by mass $m = y$

- Work done by mass m $m = mgy$
- Amount of energy converted to heat : $mgx - mgy$

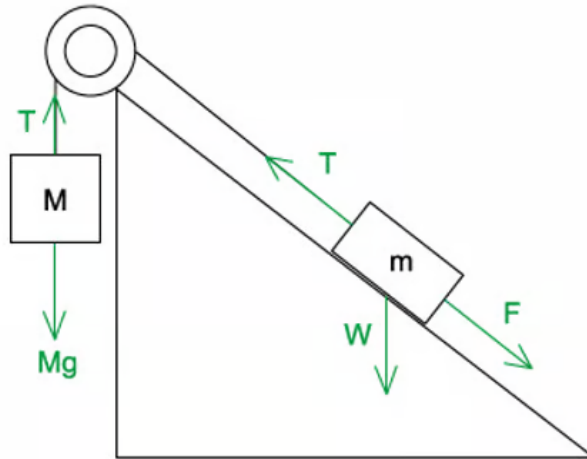


5

The correct answer is **B** because:

- The force due to the 100 N load causes the right end of the rope to accelerate downwards and simultaneously causes the left end to move upwards
- The force due to the 20 N on the force meter causes the left end to accelerate downwards and simultaneously causes the right end to move upwards
- Even if both the 100 N and the 20 N act downwards, they have opposite effects on the rope and the direction of turning of the wheel
- Since the 100 N load is greater than the 20 N force from the force meter, there is a resultant force acting downwards on the load. This force would cause the wheel to turn in a clockwise direction if not balanced by the motor
- To keep the load from moving, the wheel should also exert a force equal in magnitude and in opposite direction to the rope
 - From Newton's 3rd law since the resultant force of the rope is $100 - 20 = 80$ N downwards
- So, the force exerted by the wheel on the rope is 80 N upwards
- The wheel turns in an anticlockwise direction at a rate of 50 revolutions per

- Vertical forces on mass m : $F + W = T$
- Therefore, the frictional force is: $F = T - W = Mg - mg$
- Vertical distance travelled by mass $M = x$
- Work done by mass $M = Mgx$
- Vertical distance travelled by mass $m = y$
- Work done by mass $m = mgy$
- Amount of energy converted to heat: $mgy - Mgx$



5

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 - From Newton's 3rd law since the resultant force of the rope is $100 - 20 = 80$ N downwards
- So, the force exerted by the wheel on the rope is 80 N upwards

- The wheel turns in an anticlockwise direction at a rate of 50 revolutions per second
- This means the speed of the motor is 50 revolutions per second, and one revolution is equal to the circumference of the wheel, which is 0.3 m
- So, the distance moved against the force in one second, or its speed is $= 50 \times 0.30 = 15 \text{ m s}^{-1}$
 - Power provided by motor $= Fv = 80 \times 15 = 1200 \text{ W} = 1.2 \text{ kW}$
- Therefore, the power provided by the motor is 1.2 kW

6

The correct answer is **B** because:

- From the principle of conservation of energy:
 - The sum of energies of blocks P and Q when Q is at position B should be equal to the initial amount of gravitational potential energy (GPE) the blocks possess when Q is at position A
- When the two blocks are at rest, they are above the ground so they both possess GPE
- After Q is released, it begins to move down and transfers GPE to kinetic energy (KE) as it increases in speed, this pulls block P up the slope with the same speed and hence the same KE
- We are told the surface of the slope is rough so block P does some work against friction
- Overall, the GPE at the beginning transfers to the KE of the blocks minus the heat loss due to friction on the slope, this can be written:
 - Loss in potential energy of the whole system = Gain in kinetic energy of the whole system + work done against friction due to block P moving against a rough surface

A is incorrect because this statement is true for block Q if it was not attached to block P, but since it is you cannot ignore the energy transfers of block P

C is incorrect because this statement ignores the work done against friction as the slope is rough

D is incorrect because the loss in potential energy of the whole system is equal to (the loss in potential energy of block Q) – (gain in potential energy of block P) but this statement does not take into account the KE gained or the work done due to friction

so it is incorrect

7

The correct answer is **C** because:

- $\text{Power} = \frac{\text{work done}}{\text{time taken}} = \frac{\text{force} \times \text{distance}}{\text{time taken}}$
- Since the anchor is fixed to the pier, the boy will not be able to move it, regardless of how much force he applies, he can only move the boat towards the pier
- Hence, the work done and power will be close to zero in Diagram (a)
- For the freely floating boat Q, there is a higher chance that the boy will be able to move himself and boat Q for a distance
- A higher force will be required by the boy to travel the same distance as in Diagram (a)
- Hence, more work is done and a power output substantially larger than zero will be required in Diagram (b)
- Therefore, both work done W_b and power output P_b will be higher than work done W_a and power output P_a
 - $W_a < W_b$ and $P_a < P_b$

A is incorrect because this statement is telling us that it requires less energy and power output to pull the boat towards the pier than boat Q

B & D are incorrect because since the mass of the pier and boat Q vary greatly, the work done and power outputs to pull boat P towards them cannot be the same

8

The correct answer is **C** because:

- $\text{Power} = \frac{\text{work done}}{\text{time taken}} = \frac{\text{force} \times \text{distance}}{\text{time taken}} = \text{force} \times \text{velocity}$
- The track is frictionless, so, no frictional force acts on the mass
- If the track was horizontal the power required to give the mass m a forward acceleration a when it is moving with velocity v is given by
 - $\text{Power } P = Fv = (ma) \times v = mav \quad (1)$
- But in this case, the track is inclined at an angle θ to the horizontal
- The component of the weight of the mass (against motion) along the track is

- The component of the weight of the mass (against motion) along the track is equal to $= mg \sin \theta$
- Therefore, a force is required to keep the mass stationary and prevent it from moving down along the inclined track
- This force should be equal in magnitude to the component of the weight so that the resultant acceleration (when considering only the component of the weight and this force) is zero
- This force would result in a power equal to
 - Power $P = mg \sin \theta \times v = mgv \sin \theta$ (2)
- Total power required is given by adding Equation (1) and Equation (2):
 - Total power $= mav + mgv \sin \theta$

9

The correct answer is **D** because:

- Let's consider the motion of the ball from the point of impact with the spring to the maximum compression of the spring
- From the conservation of energy :
- Loss in kinetic energy (KE) + gravitational potential energy (GPE) = work done to compress ball = gain in elastic potential energy in spring
- Loss in ball's KE $= \frac{1}{2}mv^2$
- Gain in GPE $= mgy$
- Work done by ball = Average force $\times$ Distance $= F_{av}y$
 - $\left(\frac{1}{2}mv^2 - 0\right) + mgy = F_{av}y$
 - $F_{av} = \frac{\frac{1}{2}mv^2 + mgy}{y}$
 - $F_{av} = \frac{m}{2y}(v^2 + 2gy)$

